

EMCH 260

SOLID MECHANICS

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Torsion

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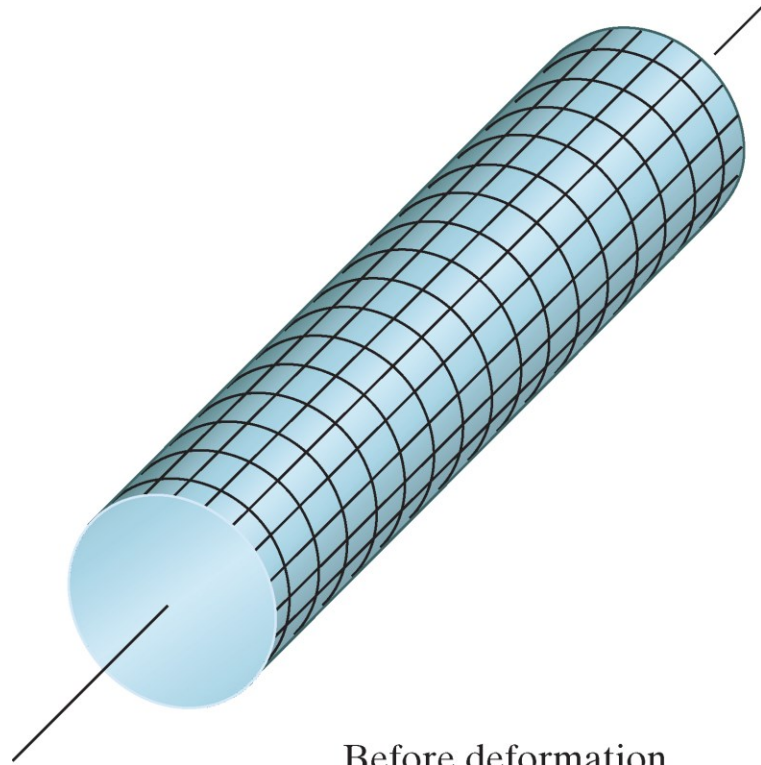
Chapter 5.1

- ✓ Understand the torsional deformation in a circular shaft due to external torque
- ✓ Understand the torsion formula and know how to use it to calculate the shear stress distribution
- ✓ Understand the physical meaning of polar moment inertia and know how to calculate it for circular cross section

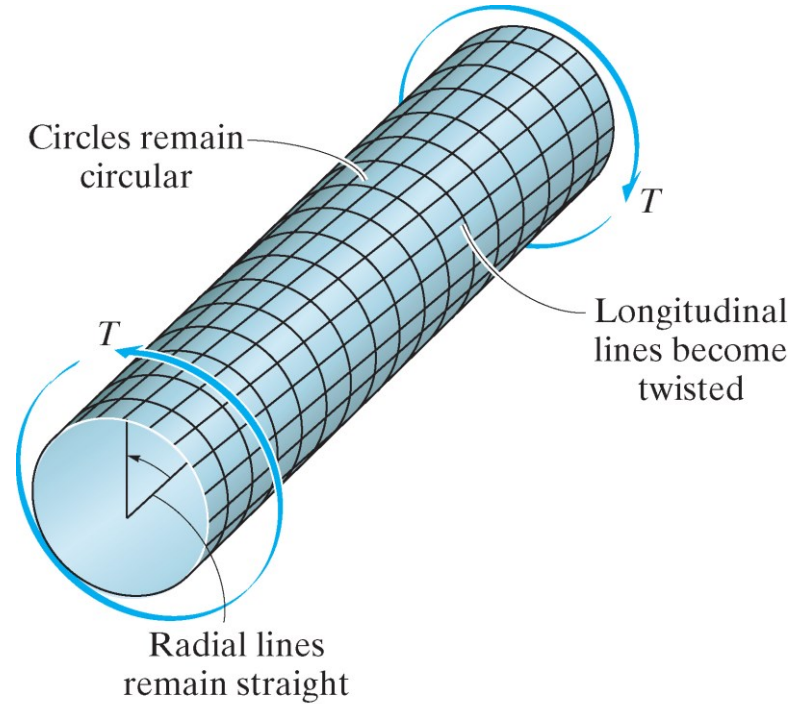
APPLICATIONS OF SEVERE TORSION



TORSIONAL DEFORMATION OF A CIRCULAR SHAFT



Before deformation
(a)

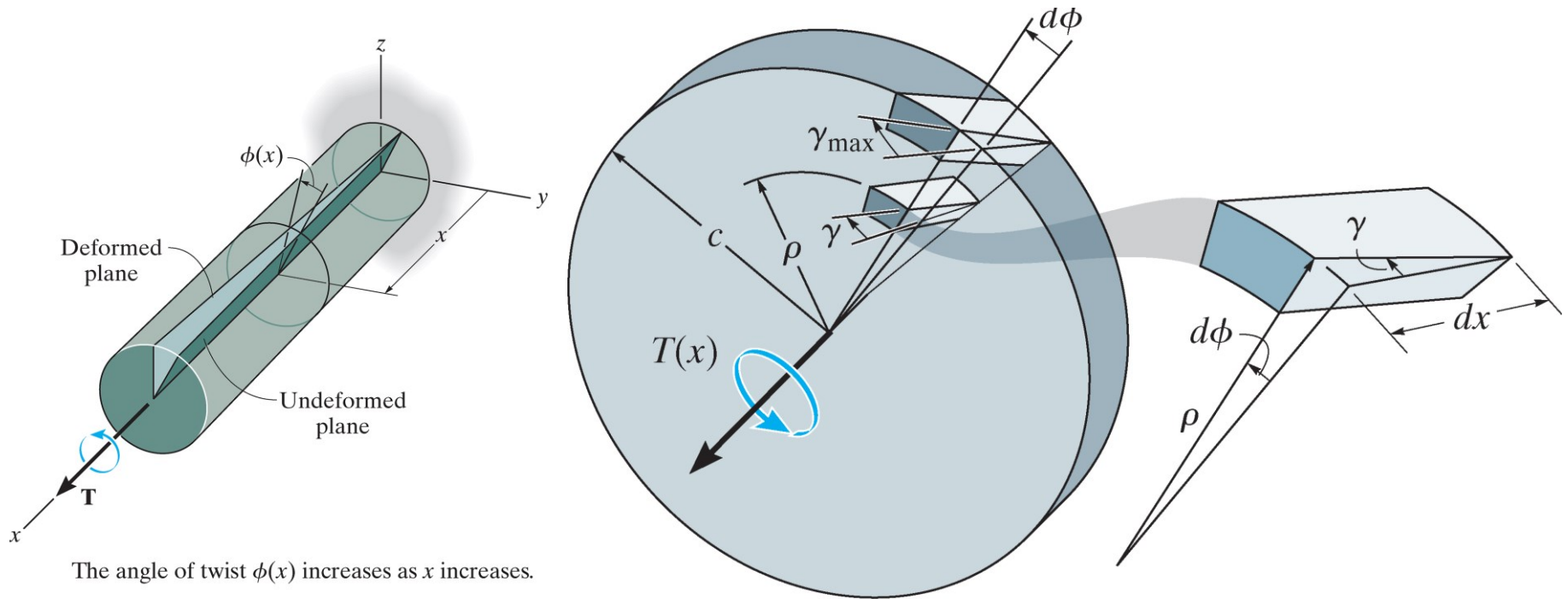


After deformation
(b)

Feature of torsional deformation

- The cross section *remains plane* while radial lines rotate
- *Shear strain and shear stress* within the material will develop due to the relative movement of adjacent cross sections

ANGLE OF TWIST AND SHEAR STRAIN



1. The shaft will rotate through an angle $\phi(x)$, called *angle of twist*, which depends on the position x and varies along the shaft axis
2. For a small disk element, the front and rear faces will undergo rotation – resulting a difference in rotation angle $d\phi$ and causing a *shear strain* (γ)

RADIAL DISTRIBUTION OF SHEAR STRAIN

$$\rho \, d\phi = dx \, \gamma$$

ρ is the radial distance from the axis of the shaft

$$\gamma = \rho \frac{d\phi}{dx}$$

$$\frac{d\phi}{dx} = \frac{\gamma}{\rho} = \frac{\gamma_{\max}}{c}$$

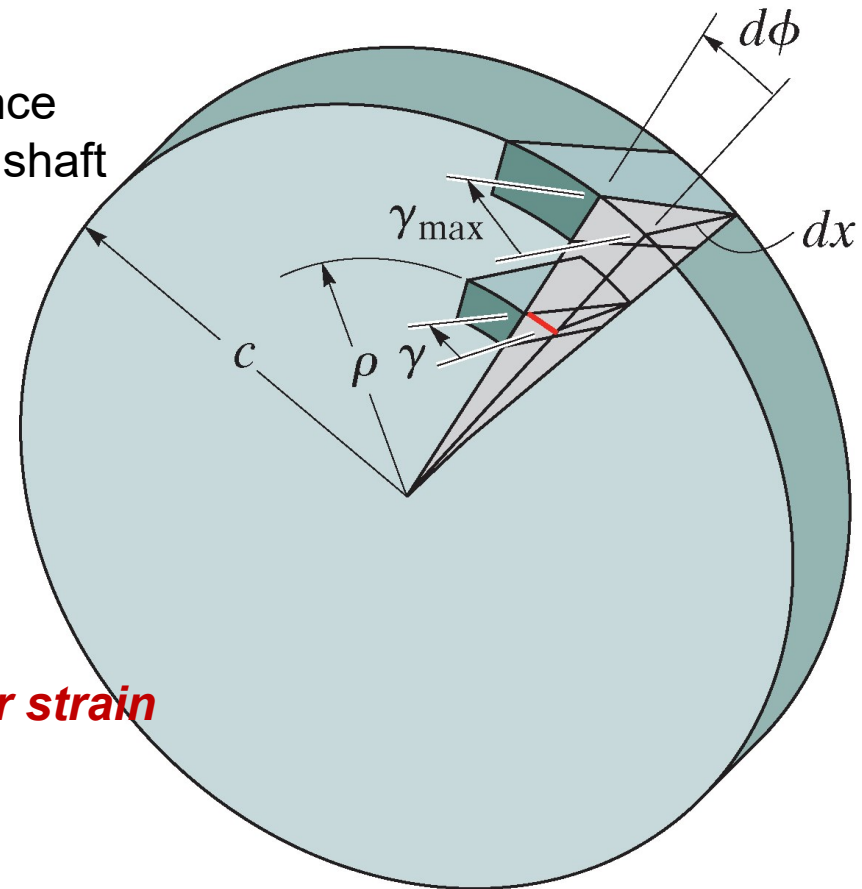
c is the radius of cross-section

Radial distribution / dependence of shear strain

$$\gamma = \left(\frac{\rho}{c}\right) \gamma_{\max}$$

or

$$\gamma(r) = \frac{r}{c} \gamma_{\max}$$



The shear strain at points on the cross section increases linearly with ρ , i.e., $\gamma = (\rho/c) \gamma_{\max}$.

TORSION FORMULA

- For linear elastic material, by applying *Hooke's law*, $\tau = G\gamma$, or $\tau_{\max} = G\gamma_{\max}$, we have the shear stress distribution

$$\tau = \left(\frac{\rho}{c}\right) \tau_{\max}$$

NOTE: Shear stress is in the plane perpendicular to the axis of the shaft

- For any cross section that is perpendicular to the axis of the shaft, the torque in the entire cross section is

$$T \equiv \int_A \rho(\tau dA) = \int_A \rho \left(\frac{\rho}{c}\right) \tau_{\max} dA$$

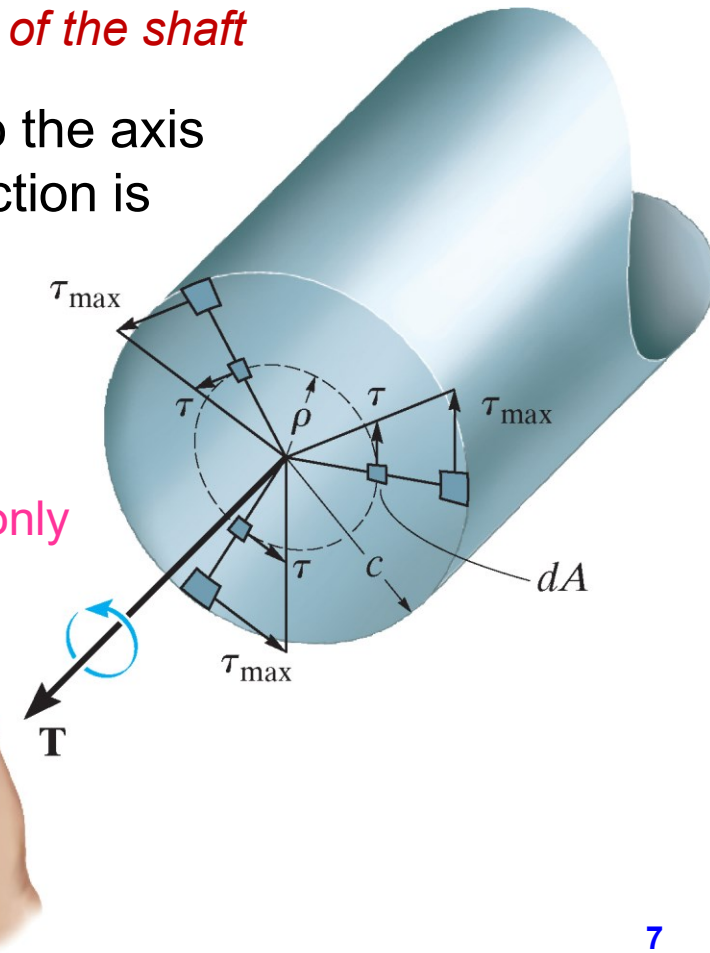
Internal resultant torque For circular cross-section only

$$\rightarrow \tau_{\max} = \frac{Tc}{J}$$

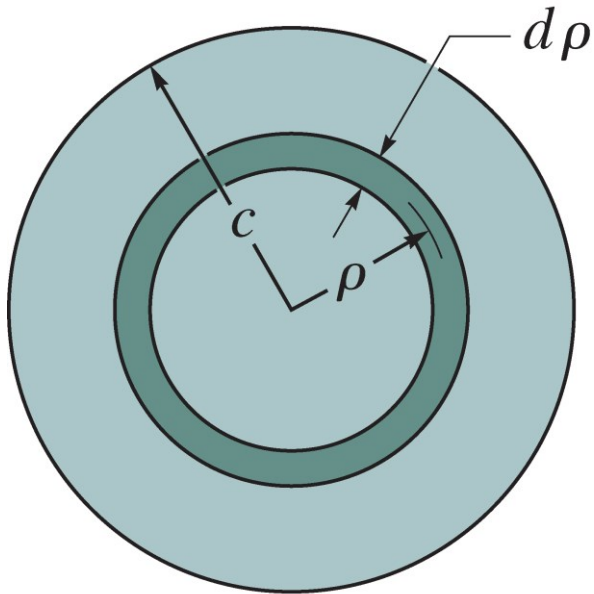
or

$$\tau = \frac{T\rho}{J}$$

J: polar moment of inertia



POLAR MOMENT OF INERTIA

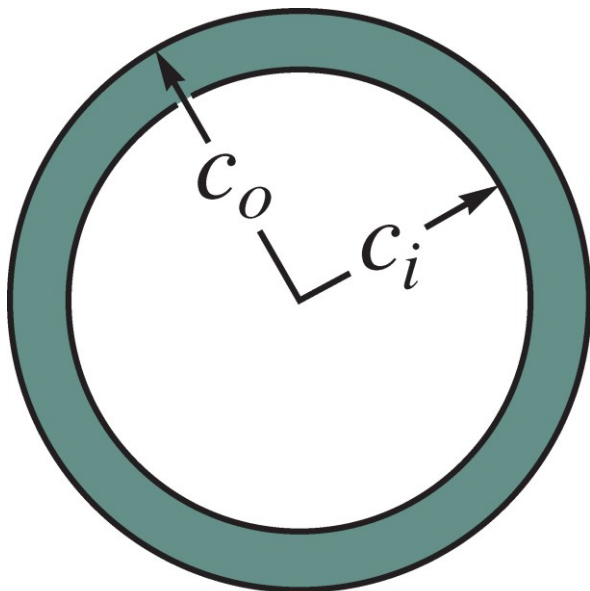


- For *solid* circular cross section

$$J = \int_A \rho^2 dA = \int_0^c \rho^2 (2\pi\rho d\rho)$$

$$J = \frac{\pi}{2} c^4$$

unit: m^4 or mm^4 or in^4

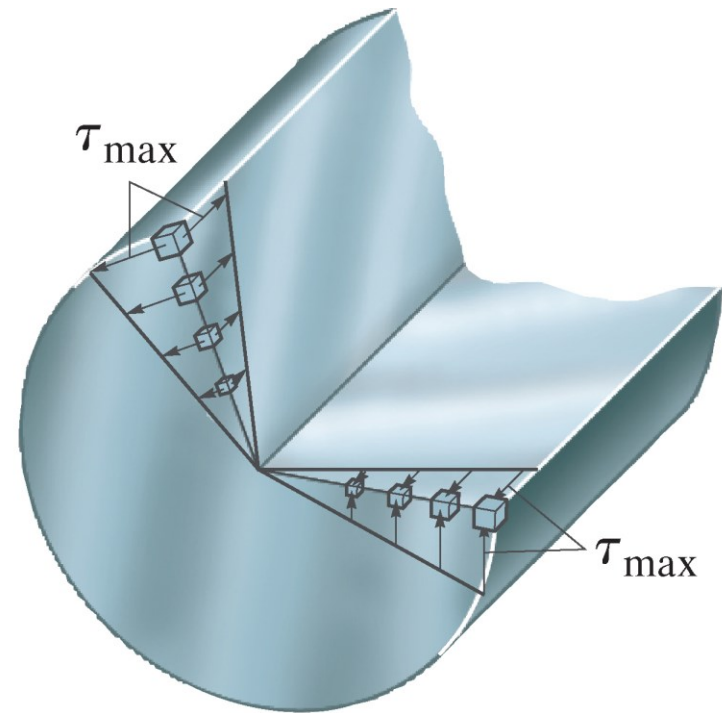
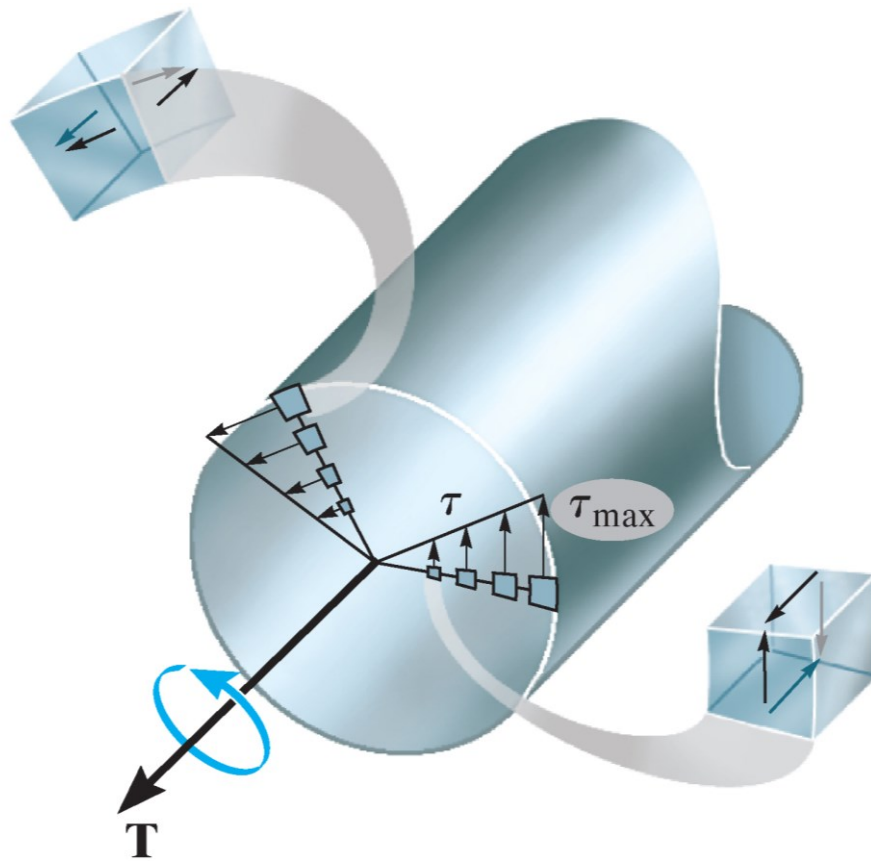


- For *hollow or tubular* cross section

$$J = \frac{\pi}{2} (c_o^4 - c_i^4)$$

Polar moment of inertia is an intrinsic geometrical properties of materials

SHEAR STRESS DISTRIBUTION



Shear stress varies linearly along each radial line of the cross section.

- Due to the complementary property of shear ([refer to Ch.1.3 p. 16](#)), the internal torque develops a linear distributing of shear stress along each radial line in the plane of the cross-sectional area, and also an associated shear-stress distribution is developed along an axial plane.

PROCEDURE FOR ANALYSIS

Step 1: Internal Torque

- Use the section method, free-body diagram, and equations of equilibrium to determine the internal torque.

Step 2: Section Property

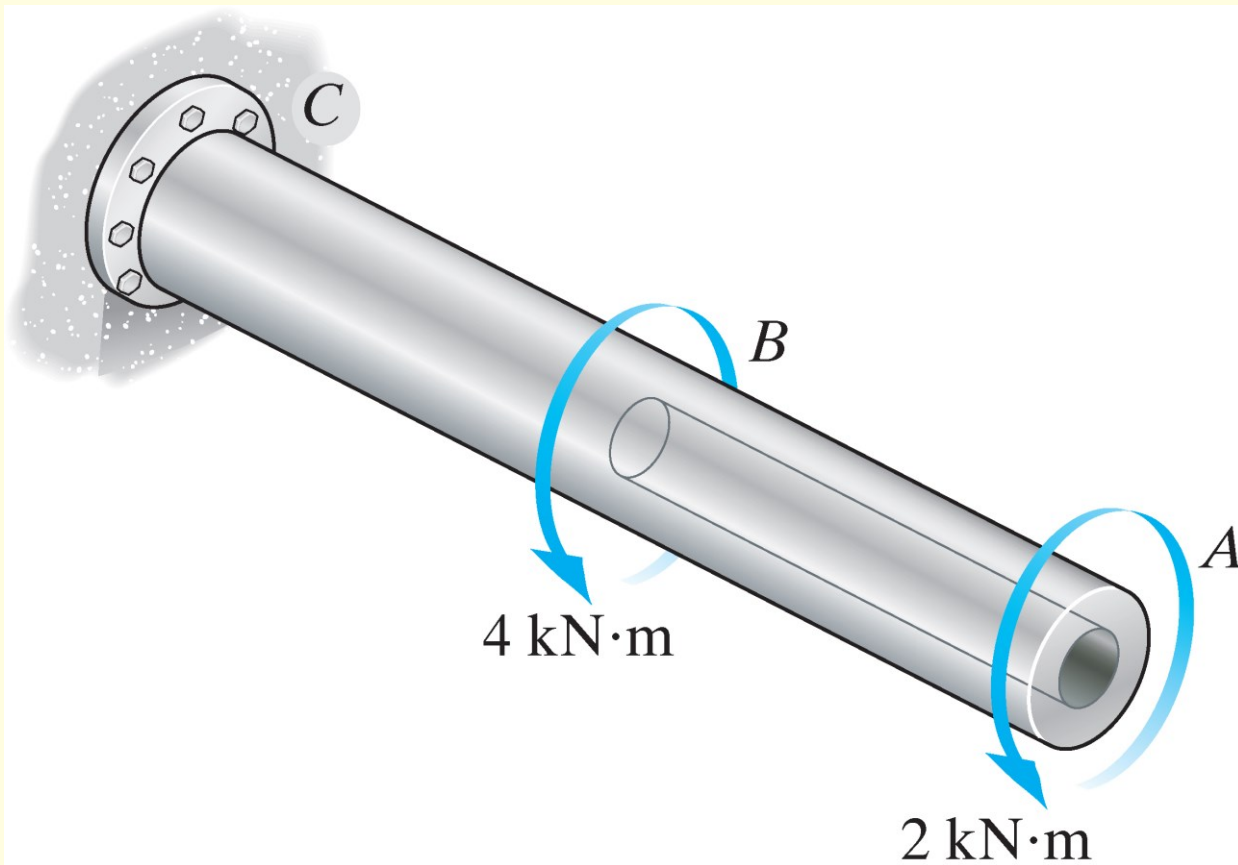
- Calculate the polar moment of inertia of the cross-sectional area.

Step 3: Shear Stress and Strain

- Apply the torsion formula to any radial distance ρ or the surface (radius equals c) to determine the shear stress (surface position corresponds to maximum shear stress)
- The shear stress acts on the cross section in a direction that is always perpendicular to ρ . The force it creates must contribute a torque about the axis of the shaft that is in the same direction as the internal resultant torque acting on the section. Once the direction is established, the direction of τ acting on all faces of element can be shown.

IN-CLASS EXAMPLE

The shaft is hollow from A to B and solid from B to C . Determine the maximum shear stress in the shaft and represent the corresponding state of stress on a volume element. The shaft has an outer diameter of 80 mm and the thickness of the wall of the hollow segment is 10 mm.

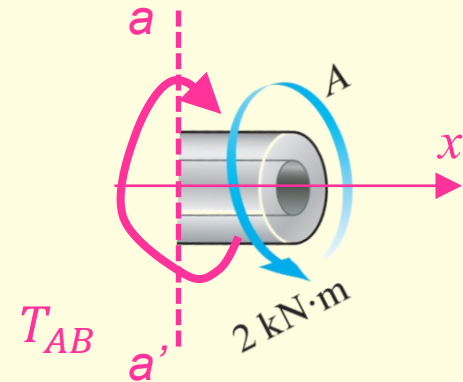
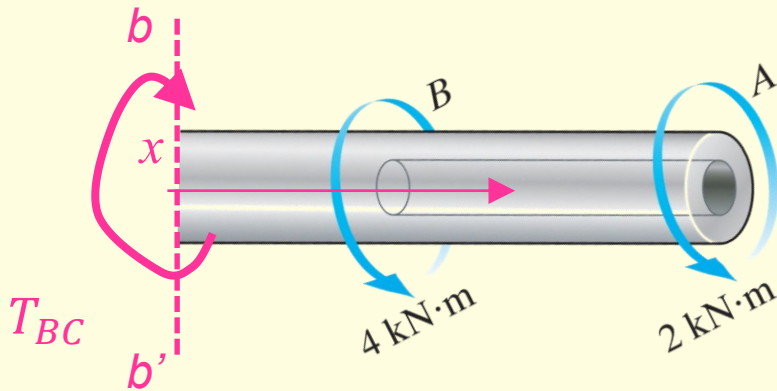


IN-CLASS EXAMPLE (cont'd)

Solutions

Step 1: Internal Torque

The external torques are applied to cross-sections A and B. Then, we have to apply free-body diagram to segment AB and BC separately to obtain the internal resultant torque.



$$T_{AB} = 2000 \text{ N}\cdot\text{m}$$

$$T_{BC} = 6000 \text{ N}\cdot\text{m}$$

IN-CLASS EXAMPLE 1 (cont'd)

Solutions

Step 2: Section Property

Polar moment of inertia of cross-section aa' and bb' is

$$J_{AB} = \frac{\pi}{2} (0.04^4 - 0.03^4) = 2.7489 \times 10^{-6} \text{ m}^4$$

$$J_{BC} = \frac{\pi}{2} (0.04^4) = 4.0212 \times 10^{-6} \text{ m}^4$$

Step 3: Shear Stress

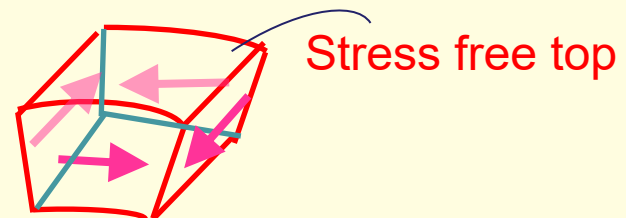
The maximum shear stress happens at the outer surface point

$$(\tau_{AB})_{\max} = \frac{T_{AB} c_{AB}}{J_{AB}} = \frac{2000 \times 0.04}{2.7489 \times 10^{-6}} = 29.10 \text{ MPa}$$

$$(\tau_{BC})_{\max} = \frac{T_{BC} c_{BC}}{J_{BC}} = \frac{6000 \times 0.04}{4.0212 \times 10^{-6}} = 59.68 \text{ MPa}$$

The maximum shear stress (59.68 MPa) occurs at the outer surface of segment BC .

state of stress on a volume element



RELEVANT PRACTICE

- **Assignment 2 on Blackboard, problem #4 (go to Blackboard under the “Assignments” folder, **Assignment 2 due: 8:30 AM, Friday, Oct. 15th 2021**)**
- **Optional Extra Exercise 2 on Blackboard, problem #3 (go to Blackboard under the “Extra Exercises” folder, **Extra Exercise 2 due: 8:30 AM, Friday, Oct. 15th 2021**)**

OUT-OF-CLASS READING

- **Textbook, pp. 183 – 193**
 - **In particular, Examples 5.1 – 5.2 (pp. 191 – 192)**